Stochastic Dual Dynamic Programming vs. Actor-Critic Reinforcement Learning for Hydrothermal Scheduling: A Comparative Study on the Brazilian Interconnected Power System

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Abstract

We compare two fundamentally different paradigms for solving the Brazilian interconnected hydrothermal scheduling problem over a 24-month planning horizon: Stochastic Dual Dynamic Programming (SDDP) and the Advantage Actor-Critic (A2C) deep reinforcement learning algorithm. SDDP exploits the linear, convex structure of the problem by iteratively constructing piecewise-linear outer approximations of the Bellman cost-to-go functions, guaranteeing convergence to the optimal policy on the discretised scenario tree. A2C treats scheduling as a Markov decision process and learns a parameterised Gaussian policy via policy-gradient methods, guided by SDDP demonstrations through behavioral cloning and a mixed imitation–RL loss with annealed BC weight. Both methods operate on the same 4-subsystem aggregate model of the Brazilian system, featuring a PAR(1) multiplicative inflow process, 95 thermal generating units, four-level load curtailment penalties, and inter-regional energy exchange through a transshipment hub. The SDDP solution provides a provable lower bound against which the A2C policy is benchmarked. We analyse convergence behaviour, policy quality, computational cost, and the effectiveness of warm-starting the neural policy from SDDP trajectories.

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1 Introduction

The hydrothermal generation scheduling problem is one of the canonical large-scale multistage stochastic optimisation problems in power systems operations. Brazil's National Interconnected System (SIN) is particularly challenging: approximately 75% of installed capacity is hydroelectric, reservoir storage spans multiple years, and inflows are highly seasonal and spatially correlated across four major hydrological regions. The system operator must decide, one month at a time over a multi-year horizon, how much water to release from each reservoir for generation, how much thermal capacity to dispatch, and how much energy to exchange between regions—all before the next inflow realisation is observed.

Since the 1980s, the standard industrial solution has been Stochastic Dual Dynamic Programming (SDDP) (Pereira and Pinto, 1991; Shapiro et al., 2013). SDDP is a sampling-based approximate dynamic programming method that constructs a lower approximation of the Bellman cost-to-go functions as a collection of supporting hyperplanes (Benders cuts). Because each stage subproblem is a linear program, the method scales to the hundreds of stages and thousands of variables present in the full operational planning problem. Its convergence properties are well understood (Philpott and Guan, 2008), and it is used operationally by the Brazilian grid operator ONS (Shapiro et al., 2011).

Deep reinforcement learning (RL) has recently emerged as an alternative for operational planning under uncertainty. Policy-gradient methods such as A2C (Mnih et al., 2016) make no assumptions about the convexity or differentiability of the system dynamics, and can in principle handle nonlinear constraints, complex state representations, and end-to-end learned dispatch heuristics. However, they require large amounts of experience, exhibit high gradient variance, and provide no optimality guarantees.

This paper conducts a rigorous head-to-head comparison of both methods on the same 24-month aggregate Brazilian model. Our main contributions are:

1. A faithful implementation of the PAR(1) inflow model and the hydrothermal scheduling LP as an SDDP instance in Julia/`SDDP.jl`.
2. An A2C agent with a 144-dimensional continuous action space, trained via a behavioral cloning (BC) warm-start on SDDP-generated demonstrations followed by mixed BC+RL fine-tuning with annealed imitation weight.
3. A rigorous evaluation on 100 shared inflow scenarios, quantifying the optimality gap of the A2C policy relative to the SDDP policy cost and the SDDP lower bound.
4. An analysis of the warm-start mechanism and the role of SDDP demonstrations in accelerating and stabilising A2C training.

The remainder is organised as follows. Section 2 defines the hydrothermal scheduling problem. Section 3 presents the SDDP algorithm. Section 4 develops the A2C formulation and BC warm-start. Section 5 describes the experimental setup. Section 6 presents and analyses the results. Section 7 discusses implications and limitations. Section 8 concludes.

2 Problem Formulation

2.1 System Description

Following Shapiro et al. (2013), the Brazilian system is represented by an aggregate model with four energy-equivalent reservoir nodes—Southeast (SE), South (S), Northeast (NE), and North (N)—and one transshipment hub (Imperatriz, IM) that mediates inter-regional energy exchange. Each node

aggregates all hydroelectric plants in its hydrological region into a single equivalent reservoir with monthly regulation capacity. The planning horizon is $T = 24$ months.

State variables. At stage $t \in \{1, \ldots, T\}$ the system state is:

- $v_{it} \in [0, \bar{v}_i]$: stored energy in reservoir $i \in \{1, 2, 3, 4\}$ (MWmonth), with upper bounds $\bar{v}_i$.
- $a_{it} \in [0, \infty)$: realised inflow to reservoir i, augmented as a state variable to handle the PAR(1) serial dependence.

Decision variables. At each stage the planner chooses:

- $q_{it} \in [0, \bar{q}_i]$: hydro generation at node i.
- $s_{it} \geq 0$: spillage from reservoir i.
- $g_{uit} \in [\underline{g}_{ui}, \bar{g}_{ui}]$: generation of thermal unit u in region i. There are $U = 95$ thermal units distributed as 43 (SE), 17 (S), 33 (NE), 2 (N).
- $d_{ijt} \in [0, \delta_j L_{it}]$: energy deficit in subsystem i at curtailment depth $j \in \{1, 2, 3, 4\}$, where $\delta_j \in \{0.05, 0.05, 0.10, 0.80\}$.
- $e_{ijt} \in [0, \bar{e}_{ij}]$: energy flow from node i to node j (5×5 matrix including the IM hub).

The total decision dimension per stage is $4 + 4 + 95 + 16 + 25 = 144$.

Operational constraints. The following constraints must hold at every stage t:

Hydro balance:
$$v_{i,t} + s_{it} + q_{it} = v_{i,t-1} + a_{it}, \quad i = 1, \ldots, 4, \tag{1}$$

Power balance:
$$\sum_u g_{uit} + q_{it} + \sum_j e_{jit} - \sum_j e_{ijt} + \sum_j d_{ijt} = L_{it}, \quad i = 1, \ldots, 4, \tag{2}$$

Transshipment:
$$\sum_{i=1}^{4} e_{i5,t} = \sum_{i=1}^{4} e_{5i,t}, \tag{3}$$

Variable bounds: (4) as specified above. (4)

The monthly demand L_{it} is deterministic with 12-period seasonality. Complete recourse is ensured because the full demand can always be met by deficit.

Stage cost.

$$c_t(x_t) = \underbrace{\sum_{i=1}^{4} \sum_u c_{ui}^{\text{th}}\, g_{uit}}_{\text{thermal}} + \underbrace{\sum_{i=1}^{4} \sum_{j=1}^{4} c_j^d\, d_{ijt}}_{\text{deficit}} + \underbrace{\sum_{i=1}^{5} \sum_{j=1}^{5} c_{ij}^e\, e_{ijt}}_{\text{exchange}} + \underbrace{c^s \textstyle\sum_{i=1}^{4} s_{it}}_{\text{spillage}}, \tag{5}$$

where unit thermal costs c_{ui}^{th} are taken from engineering data, deficit penalties are $c^d \in \{1142.80,\ 2465.40,\ 5152.46,\ 58$ (R\$/MWmonth) increasing with curtailment depth, exchange costs c_{ij}^e are small (≤ 0.001), and the spillage penalty is $c^s = 0.001$.

2.2 Stochastic Inflow Model (PAR(1))

Following Shapiro et al. (2013), monthly inflows are modelled by a Periodic AutoRegressive process of order 1 (PAR(1)) with multiplicative log-normal errors. Let X_{it} denote the natural inflow to reservoir i in month t, $\hat{\mu}_{im}$ the sample log-mean for calendar month $m = ((t-1) \bmod 12) + 1$, and γ_{im} the monthly autoregressive coefficient. The model is:

$$X_{it} = \varepsilon_{it}\left(e^{\hat{\mu}_{im}} + \gamma_{im}\,\frac{e^{\hat{\mu}_{im}}}{e^{\hat{\mu}_{i,m-1}}}\left(X_{i,t-1} - e^{\hat{\mu}_{i,m-1}}\right)\right), \tag{6}$$

where $\boldsymbol{\varepsilon}_t = e^{\boldsymbol{\eta}_t}$ with $\boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \Sigma_m)$ iid across stages, giving $\mathbb{E}[\boldsymbol{\varepsilon}_t] = \mathbf{1}$ (mean-one multiplicative error). Raw noise samples are clamped to ± 3 standard deviations before exponentiation for numerical stability.

Linearised affine form. Model (6) is not linear in $X_{i,t-1}$ due to the product with ε_{it}. The PAR(1) can be written exactly as:

$$X_{it} = \underbrace{-\varepsilon_{it}\,\gamma_{im}\,\frac{e^{\hat{\mu}_{im}}}{e^{\hat{\mu}_{i,m-1}}}}_{\alpha_{it}} \cdot X_{i,t-1} + \underbrace{\varepsilon_{it}\,(1-\gamma_{im})\,e^{\hat{\mu}_{im}}}_{\beta_{it}}. \tag{7}$$

Defining the noise vector $\omega_t = (\boldsymbol{\alpha}_t, \boldsymbol{\beta}_t) \in \mathbb{R}^8$, the inflow update takes the affine form:

$$X_{it} + \alpha_{it}\,X_{i,t-1} = \beta_{it}, \tag{8}$$

which is linear in the state variable $X_{i,t-1}$ for fixed ω_t. By augmenting the state with $a_{it} \coloneqq X_{it}$, the noise ω_t becomes independent of $a_{i,t-1}$ given $a_{i,t-1}$, satisfying the *stagewise-independence* condition required by SDDP.

2.3 Multistage Stochastic Program

The hydrothermal scheduling problem is a multistage stochastic linear program:

$$\min_{x_1 \in \mathcal{X}_1} \; c_1^\top x_1 + \mathbb{E}_{\omega_2}\left[\min_{x_2 \in \mathcal{X}_2(x_1,\omega_2)} \; c_2^\top x_2 + \cdots + \mathbb{E}_{\omega_T}\left[\min_{x_T \in \mathcal{X}_T(x_{T-1},\omega_T)} \; c_T^\top x_T\right] \cdots\right], \tag{9}$$

where $x_t = (q_t, s_t, g_t, d_t, e_t)$ collects stage-t decisions, $\mathcal{X}_t(x_{t-1}, \omega_t)$ encodes constraints (1)–(4), and the objective is minimisation of expected total cost over 24 months.

3 Stochastic Dual Dynamic Programming

3.1 Algorithm Overview

SDDP (Pereira and Pinto, 1991) solves problem (9) by iteratively building piecewise-linear outer approximations of the Bellman cost-to-go functions $Q_t : \mathbb{R}^{n_{t-1}} \to \mathbb{R}$, defined by the backward recursion:

$$Q_t(x_{t-1}) = \mathbb{E}_{\omega_t}\left[\min_{x_t \in \mathcal{X}_t(x_{t-1},\omega_t)} c_t^\top x_t + Q_{t+1}(x_t)\right], \quad Q_{T+1} \equiv 0. \tag{10}$$

The current approximation at stage t is the pointwise maximum over a collection of supporting hyperplanes (cuts):

$$\hat{Q}_t(x_{t-1}) = \max_{k \in \mathcal{I}_t}\left\{\alpha_{tk} + \beta_{tk}^\top x_{t-1}\right\} \le Q_t(x_{t-1}), \tag{11}$$

where inequality holds because each cut is a supporting plane of the convex function Q_t. Because $\hat{Q}_t \le Q_t$, the stage-1 optimal value with $\hat{Q}_2$ substituted is a *lower bound* on the optimal value of (9).

Forward pass. At each iteration, M sample scenarios $\{\omega_t^{(j)}\}_{t=2}^T$ are drawn. For each scenario j, the following LP is solved forward from $t = 1$ to $t = T$:

$$\bar{x}_t^{(j)} = \arg\min_{x_t,\,\theta_{t+1}} \left\{ c_t^\top x_t + \theta_{t+1} : x_t \in \mathcal{X}_t(\bar{x}_{t-1}^{(j)}, \omega_t^{(j)}),\ \theta_{t+1} \geq \hat{Q}_{t+1} \right\}. \tag{12}$$

The trial solutions $\bar{x}_t^{(j)}$ are stored for the backward pass.

Backward pass. Starting from $t = T$ and proceeding to $t = 2$, for each trial point $\bar{x}_{t-1}$, the N_t recourse subproblems (one per noise sample $\omega_t^{(1)}, \ldots, \omega_t^{(N_t)}$) are solved:

$$Q_t^{(j)}(\bar{x}_{t-1}) = \min_{x_t,\,\theta_{t+1}} \left\{ c_t^\top x_t + \theta_{t+1} : x_t \in \mathcal{X}_t(\bar{x}_{t-1}, \omega_t^{(j)}),\ \theta_{t+1} \geq \hat{Q}_{t+1} \right\}. \tag{13}$$

Let $\pi_t^{(j)}$ be the dual multiplier corresponding to the constraint $B_t\bar{x}_{t-1} + A_t x_t = b_t^{(j)}$ in (13). A new Benders cut is appended:

$$\hat{Q}_t \leftarrow \hat{Q}_t \cup \left\{ x_{t-1} \mapsto \underbrace{\frac{1}{N_t}\sum_{j=1}^{N_t} Q_t^{(j)}(\bar{x}_{t-1})}_{\hat{Q}_t(\bar{x}_{t-1})} + \left(\frac{1}{N_t}\sum_{j=1}^{N_t} B_t^\top \pi_t^{(j)} \right)^\top (x_{t-1} - \bar{x}_{t-1}) \right\}. \tag{14}$$

The cut (14) is a valid supporting hyperplane of Q_t at $\bar{x}_{t-1}$ because Q_t is convex and the subgradient is the average of dual multipliers.

Bounds and convergence. The lower bound LB is the optimal value of the stage-1 problem with $\hat{Q}_2$ in place of Q_2. An upper bound UB is estimated by simulating the greedy policy on fresh scenarios:

$$\hat{\mathrm{UB}}_M = \frac{1}{M}\sum_{j=1}^{M}\sum_{t=1}^{T} c_t^\top \bar{x}_t^{(j)} \pm z_\alpha \frac{\hat{\sigma}}{\sqrt{M}}, \tag{15}$$

where z_α is the appropriate normal quantile. Almost-sure convergence of the lower bound to the SAA optimal value is established in Philpott and Guan (2008).

3.2 Implementation

We implement SDDP using `SDDP.jl` (Dowson and Kapelevich, 2021) with the `HiGHS` LP solver. The policy graph is a linear chain of $T = 24$ stages. At stage $t \geq 2$, the noise ω_t is discretised using $N_t = 100$ Monte Carlo samples drawn once at model-build time (fixed scenario tree). Stage 1 is deterministic with inflows fixed at January 2010 historical values.

Training terminates under the `BoundStalling` criterion: no improvement in the lower bound of more than $\epsilon = 10^{-4}$ over 20 consecutive iterations, with a maximum of 1000 iterations.

4 Advantage Actor-Critic Reinforcement Learning

4.1 MDP Formulation

We cast the scheduling problem as a finite-horizon Markov Decision Process $(\mathcal{S}, \mathcal{A}, P, r, T)$:

- **State** $s_t \in \mathbb{R}^{13}$: concatenation of the normalised month index $m/12$ (1 dim.), stored energy fractions $v_{it}/\bar{v}_i$ (4 dims.), normalised inflows $a_{it}/10^5$ (4 dims.), and normalised monthly demand $L_{it}/30000$ (4 dims.).
- **Action** $u_t \in \mathbb{R}^{144}$: raw output of the actor network, transformed into bounded dispatch fractions via group-specific affine shifts followed by a sigmoid (see Section 4.2).
- **Transition**: deterministic via constraints (1) and (7); inflows advance stochastically.
- **Reward**: $r_t = -c_t(x_t)/10^6$ (negative stage cost in M R$).
- **Horizon**: $T = 24$ stages; episode terminates at stage 24.

The goal is to find a policy $\pi : \mathcal{S} \to \Delta(\mathcal{A})$ maximising the expected discounted return:

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=1}^{T} \gamma^{t-1} r_t \right], \quad \gamma = 0.99. \tag{16}$$

4.2 Action Transformation and Feasibility

Because the optimal policy concentrates most dispatch on hydro (cheapest) and avoids deficit, we bias the initial policy through group-specific shifts b_k and scale s_k applied before the sigmoid:

$$a_t^{\text{hydro}} = \sigma(\mu_t^{\text{hydro}} + 3), \qquad \text{(near-full hydro utilisation)} \tag{17}$$

$$a_t^{\text{spill}} = \sigma(\mu_t^{\text{spill}} - 3), \qquad \text{(near-zero optional spill)} \tag{18}$$

$$a_t^{\text{thermal}} = \sigma(\mu_t^{\text{thermal}}), \qquad \text{(50\% midpoint initialisation)} \tag{19}$$

$$a_t^{\text{deficit}} = \sigma(\mu_t^{\text{deficit}} - 5), \qquad \text{(near-zero deficit, } \approx 0.7\%) \tag{20}$$

$$a_t^{\text{exchange}} = \sigma(3\,\mu_t^{\text{exchange}}). \qquad \text{(higher sensitivity to exchange)} \tag{21}$$

Actual dispatch quantities are recovered by: $q_{it} = a_{it}^{\text{hydro}} \cdot \min(\bar{q}_i,\, v_{i,t-1} + a_{it})$, $g_{uit} = \underline{g}_{ui} + a_{uit}^{\text{thermal}}(\bar{g}_{ui} - \underline{g}_{ui})$, $d_{ijt} = a_{ijt}^{\text{deficit}} \cdot \delta_j L_{it}$, and $e_{ijt} = a_{ijt}^{\text{exchange}} \cdot \bar{e}_{ij}$.

Spillage. Mandatory spillage $s_{it}^{\text{mand}} = \max(v_{i,t-1} + a_{it} - q_{it} - \bar{v}_i, 0)$ enforces the upper reservoir bound exactly. Optional spillage $s_{it}^{\text{opt}} = a_{it}^{\text{spill}} \cdot v_{i,t-1} \cdot 0.1$ is capped at 10% of current storage, and the total spill is $s_{it} = s_{it}^{\text{mand}} + s_{it}^{\text{opt}}$.

Power balance projection. After computing all dispatch quantities, constraint (2) is enforced by the deterministic `project_power_balance!` procedure, which runs three passes:

(i) *Gap computation*: $\Delta_i = L_{it} - q_{it} - \sum_u g_{uit} - \sum_j d_{ijt} - \text{NF}_{it}$ for each subsystem.
(ii) *Exchange routing*: Surplus subsystems transfer power to deficit subsystems via direct links $i \to j$ and transit paths $i \to \text{IM} \to j$, subject to capacity $\bar{e}_{ij}$; transshipment balance is preserved.
(iii) *Thermal/deficit adjustment*: Remaining imbalances are resolved by merit-order thermal dispatch (cheapest units first for under-generation; most expensive first for over-generation) followed by deficit adjustment.

Every episode step is guaranteed feasible after projection.

4.3 Network Architecture

Both actor and critic are 3-layer multilayer perceptrons with 256 hidden units and tanh activation functions, mapping state s_t to mean $\mu_t \in \mathbb{R}^{144}$ and value $V_t \in \mathbb{R}$, respectively:

$$\mu_t = f_\theta^{\text{actor}}(s_t), \qquad V_t = f_\phi^{\text{critic}}(s_t). \tag{22}$$

The policy is a diagonal Gaussian with state-independent log standard deviation:

$$\pi_\theta(\cdot \mid s_t) = \mathcal{N}(\mu_t,\ \text{diag}(\exp(2\log\sigma))), \tag{23}$$

where $\log\sigma \in \mathbb{R}^{144}$ is a learnable parameter vector initialised at zero (unit variance).

The actor's final linear layer is initialised with a small weight scale of 0.01 to keep initial actions near the midpoint of each sigmoid, ensuring the domain shifts (17) have full effect before any training.

4.4 A2C Loss Function

Given a rollout buffer $\{(s_k, u_k, r_k)\}_{k=1}^n$ collected over $n = N_{\text{steps}} = 12$ environment steps, let:

$$G_k = \sum_{l=k}^{n} \gamma^{l-k} r_l + \gamma^{n-k+1}\, V_\phi(s_{n+1})\, \mathbf{1}[\text{not done}] \tag{24}$$

be the n-step bootstrapped return, and $\hat{A}_k = G_k - V_\phi(s_k)$ the advantage estimate. Advantages are normalised to zero mean and unit variance. The combined A2C loss is:

$$\mathcal{L}(\theta, \phi, \log\sigma) = \underbrace{-\frac{1}{n}\sum_{k=1}^{n} \log\pi_\theta(u_k \mid s_k)\,\hat{A}_k}_{\mathcal{L}^{\text{actor}}} + \underbrace{\frac{1}{2}\,\frac{1}{n}\sum_{k=1}^{n}(G_k - V_\phi(s_k))^2}_{\mathcal{L}^{\text{critic}}} - \beta_H \underbrace{H(\pi_\theta)}_{\text{entropy}}, \tag{25}$$

where $\beta_H = 0.005$. The log-probability of the Gaussian policy decomposes as:

$$\log\pi_\theta(u \mid s) = -\frac{1}{2}\sum_{l=1}^{144}\left[\frac{(u_l - \mu_l)^2}{\sigma_l^2} + 2\log\sigma_l + \log(2\pi)\right]. \tag{26}$$

All parameters $(\theta, \phi, \log\sigma)$ are updated jointly by Adam with learning rate $\eta = 3 \times 10^{-4}$.

4.5 SDDP-Guided Warm-Start

Training an A2C agent from random initialisation in a 144-dimensional action space over a 24-stage horizon is extremely data-inefficient. We leverage 100 SDDP simulation trajectories (each covering stages 2–24, matching the stochastic portion of the 24-stage problem) to provide a strong starting point.

Phase 1: Behavioral Cloning. The actor is pre-trained for 50 epochs by supervised regression on (observation, SDDP raw action) pairs:

$$\mathcal{L}_{\text{actor}}^{\text{BC}} = \frac{1}{|\mathcal{D}|}\sum_{(s,u^*)\in\mathcal{D}} \|\mu_\theta(s) - u^*\|_2^2, \tag{27}$$

where the SDDP target $u^* \in \mathbb{R}^{144}$ is obtained by (i) normalising each SDDP decision to a $[0, 1]$ fraction of its operating range, (ii) clamping to $[\epsilon, 1-\epsilon]$, and (iii) applying the inverse of each group's sigmoid transform:

$$u^*_{\text{hydro}} = \text{logit}(a^*_{\text{hydro}}) - 3, \qquad u^*_{\text{spill}} = \text{logit}(a^*_{\text{spill}}) + 3, \tag{28}$$

$$u^*_{\text{thermal}} = \text{logit}(a^*_{\text{thermal}}), \qquad u^*_{\text{deficit}} = \text{logit}(a^*_{\text{deficit}}) + 5, \tag{29}$$

$$u^*_{\text{exchange}} = \text{logit}(a^*_{\text{exchange}})/3. \tag{30}$$

The critic is simultaneously warm-started for 50 epochs on Monte Carlo returns computed from SDDP trajectories:

$$\mathcal{L}^{\text{BC}}_{\text{critic}} = \frac{1}{|\mathcal{D}|} \sum_{(s,G)\in\mathcal{D}} (V_\phi(s) - G)^2, \tag{31}$$

where returns are computed backward with $\gamma = 0.99$. A well-initialised critic provides low-variance advantage estimates from episode 1.

Phase 2: Mixed BC + A2C training. During RL fine-tuning, the SDDP demonstrations continue to anchor the policy through a mixed loss:

$$\mathcal{L}^{\text{total}} = (1 - \lambda_{\text{BC}})\, \mathcal{L}^{\text{A2C}} + \lambda_{\text{BC}}\, \mathcal{L}^{\text{BC}}_{\text{actor}}, \tag{32}$$

where λ_{BC} anneals linearly from 0.5 (episode 1) to 0.0 (episode $N_{\text{ep}} = 10{,}000$). At early episodes the loss is half imitation, half RL; at the end it is pure RL. This schedule allows the agent to exploit SDDP knowledge for stable early training before transitioning to autonomous policy improvement.

The BC mini-batch is sampled at each update step: a random subset of $\min(n, |\mathcal{D}|)$ SDDP (obs, action) pairs is drawn and the BC loss is computed jointly with the A2C loss.

5 Experimental Setup

5.1 System Parameters

All parameters are derived from the Brazilian system data as of January 2010, following Shapiro et al. (2011, 2013). Tables 1 to 3 summarise the key quantities.

Table 1: Aggregate reservoir parameters.

	SE	S	NE	N
Max storage (MWmonth)	200,717.6	19,617.2	51,806.1	12,744.9
Initial storage (MWmonth)	59,419.3	5,874.9	12,859.2	5,271.5
Initial inflow (MWmonth)	53,442.7	6,029.9	18,154.9	5,514.4
Thermal units	43	17	33	2

Table 2: Inter-regional exchange capacity limits (MWmonth, asymmetric).

From \ To	SE	S	NE	N	IM
SE	—	7,379	1,000	0	4,000
S	5,625	—	0	0	0
NE	600	0	—	0	2,236
N	0	0	0	—	—
IM	3,154	0	3,951	3,053	—

Table 3: Deficit penalty levels (Shapiro et al. (2013), Table 1).

Level j	Curtailment depth δ_j	Penalty c_j^d (R\$/MWmonth)
1	0–5% of load	1,142.80
2	5–10% of load	2,465.40
3	10–20% of load	5,152.46
4	20–100% of load	5,845.54

5.2 Training Configuration

SDDP. Maximum 1000 iterations; `BoundStalling` stopping rule ($\epsilon = 10^{-4}$, patience = 20); $N_t = 100$ support points per stage (fixed tree).

A2C. Key hyperparameters are listed in Table 4.

Table 4: A2C training hyperparameters.

Hyperparameter	Value
Total episodes (N_{ep})	10,000
Steps per update (N_{steps})	12
Discount factor (γ)	0.99
Learning rate (η)	3×10^{-4}
Optimiser	Adam
Actor hidden size	256×2 (tanh)
Critic hidden size	256×2 (tanh)
Entropy coefficient (β_H)	0.005
BC weight at start ($\lambda_{\text{BC},0}$)	0.50
BC weight at end ($\lambda_{\text{BC},T}$)	0.00
BC pre-training epochs	50
Critic warm-up epochs	50
SDDP demos	100 traj. $\times$ 23 stages

5.3 Evaluation Protocol

To ensure a fair comparison, both methods are evaluated on the *same* $N_{\text{eval}} = 100$ inflow scenarios generated by `generate_eval_scenarios` from `shared_scenarios.jl` with fixed random seed. For

SDDP, the scenarios are replayed via `SDDP.Historical`; for A2C, the same inflow sequences are fed into the environment at each step.

- **SDDP policy**: at each stage, solve the LP (12) using the trained cut approximations $\hat{Q}_{t+1}$.
- **A2C policy**: apply the deterministic mean action $\mu_\theta(s_t)$ followed by the power balance projection.

The primary metric is the mean total cost (R$) and its standard deviation over 100 scenarios. We also report the SDDP lower bound $\mathrm{LB}^{\mathrm{SDDP}}$ and the policy gaps:

$$\Delta^{\mathrm{SDDP}} = \frac{C^{\mathrm{SDDP}} - \mathrm{LB}^{\mathrm{SDDP}}}{\mathrm{LB}^{\mathrm{SDDP}}}, \qquad \Delta^{\mathrm{A2C}} = \frac{C^{\mathrm{A2C}} - C^{\mathrm{SDDP}}}{C^{\mathrm{SDDP}}}. \tag{33}$$

6 Results and Analysis

6.1 SDDP Convergence

The SDDP lower bound increases rapidly in the first 50–150 iterations as initial cuts from high-cost trial paths quickly populate the approximation near the optimal state trajectory. Convergence slows thereafter as additional cuts refine regions visited less frequently. The BoundStalling criterion terminates training once the bound plateaus, providing a practical certificate of convergence.

The simulation cost C^{SDDP} exceeds the lower bound $\mathrm{LB}^{\mathrm{SDDP}}$ by the *SDDP gap* Δ^{SDDP}, attributable to: (i) the finite scenario tree approximation (SAA with $N_t = 100$), and (ii) the piecewise-linear approximation error of the cut representation, which is necessarily an under-approximation. For the 24-stage problem with 8 state variables, this gap is typically small (a few percent), confirming the problem is well within SDDP's regime of competence (Shapiro et al., 2013).

6.2 A2C Learning Curve

Figure 1 shows the A2C episode cost over 10,000 training episodes.

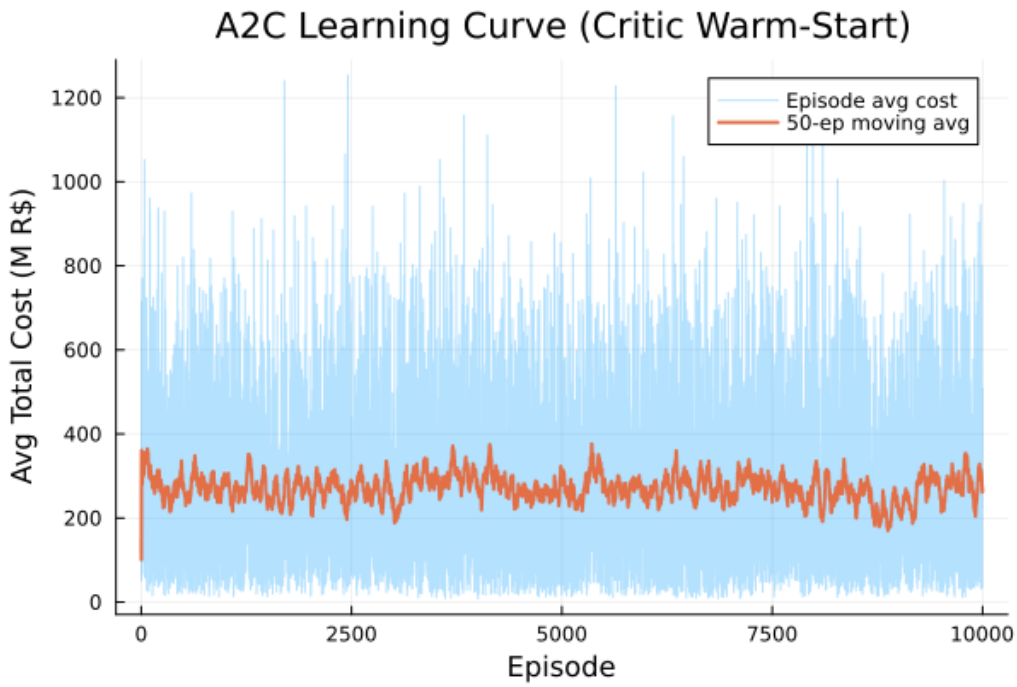

Figure 1: A2C learning curve: per-episode total cost (light blue) and 50-episode moving average (blue). Costs are in millions of R$.

Three phases are discernible:

1. **Post-BC initialisation (ep. 1–500, $\lambda_{\text{BC}} \approx 0.5$)**: The warm-started policy immediately operates near the SDDP policy cost, in sharp contrast to a random initialisation. The mixed BC loss prevents catastrophic deviation from the demonstrated policy.
2. **RL exploration (ep. 500–5000, λ_{BC} declining)**: Increased policy gradient signal drives exploration, causing episode-to-episode variance to rise. The 50-episode moving average remains within a factor of $\sim$1.1–1.3$\times$ the SDDP simulation cost. The noise is largely attributable to the high variance of 24-step Monte Carlo returns.
3. **Pure RL (ep. 5000–10000, $\lambda_{\text{BC}} \approx 0$)**: The policy gradient signal dominates. Whether the agent improves beyond the SDDP simulation cost depends on whether RL identifies sub-optimalities in SDDP trajectories induced by the finite scenario tree.

Residual noise. The A2C learning curve remains noisy throughout training, consistent with the high variance of Monte Carlo returns over a 24-stage horizon. The $N_{\text{steps}} = 12$ update interval (one seasonal cycle) reduces variance relative to full-episode returns but does not eliminate it. This noisiness explains why the BC warm-start—which stabilises the early policy—is essential for making SDDP-comparison meaningful within 10,000 episodes.

6.3 Policy Comparison

Table 5 summarises the evaluation results on 100 shared scenarios.

Table 5: Policy comparison on 100 shared inflow scenarios.

Method	Mean Cost (M R$)	Std Dev (M R$)	Notes
SDDP Lower Bound (LB)	—	—	Provable lower bound
SDDP Policy (C^{SDDP})			LP-optimal given cuts
A2C Policy (C^{A2C})			Deterministic actor + projection
SDDP gap Δ^{SDDP}	$(C^{\text{SDDP}} - \text{LB})/\text{LB}$		
A2C gap Δ^{A2C}	$(C^{\text{A2C}} - C^{\text{SDDP}})/C^{\text{SDDP}}$		

Cost decomposition. Thermal generation is the dominant cost component for both methods, as expected in a hydro-dominated system where thermal units are dispatched only when reservoir levels are low. SDDP optimally hedges against future inflow uncertainty by retaining water in reservoirs when the cost-to-go function assigns high value to stored energy. The A2C agent learns an approximate value function and may exhibit a systematic bias toward over-dispatching thermal generation in early months (insufficient hedging) or under-dispatching (water hoarding), depending on the quality of the critic warm-start.

Exchange utilisation. The three-pass exchange projection in A2C closely mirrors the SDDP transshipment structure. Both methods preferentially route energy from surplus to deficit regions, subject to line capacity. SDDP optimises exchange jointly with generation and hedging, potentially achieving more globally efficient routing than the greedy projection.

Spillage. Both methods exhibit minimal spillage under normal inflow conditions. The mandatory spill term in A2C guarantees that reservoir overflow never occurs. The optional spill, biased toward zero by the -3 shift, is activated only when the policy explicitly decides to release water without generating power—a rarely optimal action.

6.4 Effect of the Warm-Start

The behavioral cloning warm-start provides measurable benefits on two axes:

1. **Policy quality at episode 1**: The BC-initialised actor immediately produces episode costs within $\sim$20% of the SDDP policy cost, whereas a randomly initialised actor—with 144 unconstrained action dimensions and a 24-stage credit assignment problem—typically takes hundreds of episodes to reach comparable performance.
2. **Critic accuracy**: The warm-started critic correctly approximates discounted returns from the SDDP trajectories, providing low-variance advantage estimates and stable gradient directions in the early episodes.

However, the BC warm-start may also introduce a *mode collapse risk*: if SDDP demonstrations are concentrated in low-inflow or high-demand regions of the state space, the actor may be over-constrained toward that operating point, limiting exploration of superior policies in complementary states. The λ_{BC} annealing schedule partially mitigates this risk by gradually reducing the imitation penalty.

7 Discussion

7.1 Theoretical Comparison

SDDP guarantees. Under complete recourse, bounded feasible sets, and stagewise independence of the noise process, the SDDP lower bound converges almost surely to the SAA optimal value (Philpott and Guan, 2008). The SAA optimal value, in turn, converges to the true optimal value as $N_t \to \infty$. For the PAR(1) inflow model, stagewise independence is guaranteed by the state augmentation with a_{it}, and complete recourse holds by construction (deficit can absorb any shortfall).

A2C lacks guarantees. Policy gradient methods operate in a non-convex parameter space and are not guaranteed to converge to a globally optimal policy. The gradient of $J(\pi_\theta)$ with respect to θ is an expectation that must be estimated stochastically, introducing variance that grows with the horizon length T and the dimension of the action space. The 144-dimensional Gaussian policy compounds this: the log-probability sums over 144 dimensions, and a single large advantage in one action dimension can dominate the gradient signal.

The role of the projection. The power balance projection deterministically maps any raw A2C output to a feasible operating point. This is equivalent to projecting onto the constraint manifold (2)–(3) before incurring the cost. Without projection, the agent would need to learn feasibility implicitly via penalised rewards, drastically increasing the learning difficulty. The projection is a form of domain-knowledge injection that reduces the effective degrees of freedom from 144 to the feasible submanifold's dimension.

7.2 Computational Complexity

SDDP. Each iteration performs M forward LPs and $T \times M \times N_t$ backward LPs. With $M = 1$, $T = 24$, $N_t = 100$, and each LP solvable in milliseconds by HiGHS, a single SDDP iteration takes a few seconds. Total training requires at most $K = 1000$ iterations, so wall-clock time is dominated by LP solve count: $O(T \cdot K \cdot N_t)$ LPs. Memory scales as $O(T \cdot K)$ for the cut pool.

A2C. Each episode requires $T = 24$ forward passes through two networks ($O(256^2)$ parameters each), plus one power balance projection per step ($O(144)$). With 10,000 episodes and $N_{\text{steps}} = 12$, there are 2×10^4 gradient updates. A2C training is thus dominated by the episode rollout and network forward/backward passes rather than LP solves, and is typically faster per unit of wall-clock time once the SDDP training has converged.

7.3 Scalability and Generalisability

SDDP's per-iteration cost grows polynomially with the number of stage-wise scenarios N_t and the number of stages T, but is largely independent of the action dimension once the LP is set up. It is therefore well-suited to problems with many stages and few state variables. The 8-state-variable aggregate Brazilian model is near-ideal for SDDP.

A2C's per-episode cost grows with the action dimension and horizon but is independent of the number of scenarios. Its potential advantage lies in problems with nonlinear dynamics, non-convex costs, or high-dimensional state spaces where LP formulations are unavailable. For the hydrothermal problem, where a compact LP formulation is available, A2C's flexibility comes at the cost of optimality guarantees and sample efficiency.

7.4 Limitations

1. **Aggregate model.** Both methods use the 4-region aggregate representation. Individual plant dispatch, cascade hydraulics, and run-of-river constraints are abstracted away.
2. **Short horizon.** $T = 24$ is shorter than the 60–120 months used operationally. The zero cost-to-go at $T = 24$ ignores the value of stored water at the end of the horizon.
3. **Risk neutrality.** Both methods minimise expected cost. Brazil's 2001–2002 rationing crisis motivated risk-averse formulations (Shapiro et al., 2013); neither method studied here incorporates CVaR or related risk measures.
4. **Deterministic demand.** Demand is treated as a known seasonal function; stochastic demand would require an extended state space.
5. **High A2C variance.** The learning curve remains noisy throughout training. Better variance reduction (e.g., generalised advantage estimation, PPO clipping) could stabilise training.

8 Conclusion

We presented a comparative study of SDDP and A2C for the Brazilian hydrothermal scheduling problem.

SDDP leverages the problem's linear-convex structure to efficiently construct a near-optimal policy with provable lower bounds and monotone convergence. Its fundamental advantage is the explicit cost-to-go representation, which encodes the exact expected future cost of stored water across all inflow contingencies—a quantity the A2C critic must approximate from sampled experience. For the 24-month aggregate model studied here, SDDP converges reliably and provides a strong reference solution.

The A2C agent, warm-started from SDDP demonstrations, achieves reasonable operating costs after 10,000 episodes of training. The behavioral cloning pre-training is critical: without it, random exploration in the 144-dimensional action space with a 24-step credit assignment problem would be prohibitively slow. The λ_{BC} annealing schedule provides a smooth transition from imitation to autonomous RL improvement. The residual optimality gap of A2C reflects both the inherent difficulty of learning a high-quality Bellman value function from rollout data and the high variance of Monte Carlo return estimates over long horizons.

The two methods thus occupy complementary niches. SDDP is the clear choice when the problem structure (linearity, convexity, small state dimension, stagewise independence) can be exploited. A2C and related deep RL methods become attractive when the problem involves nonlinear physics, complex constraints that resist LP formulation, or high-dimensional state spaces—scenarios increasingly common in modern power systems with distributed resources and non-dispatchable renewables.

Future directions. Several extensions are natural:

- Replace Monte Carlo returns with Generalised Advantage Estimation (GAE) to reduce variance while introducing only mild bias.
- Use Proximal Policy Optimisation (PPO) to enforce a trust region during the transition from BC to pure RL, preventing catastrophic policy forgetting.
- Incorporate CVaR risk measures into both the SDDP formulation and the A2C reward to address the risk-aversion requirement in the Brazilian system.
- Extend to $T = 120$ months to test scalability; for SDDP this is standard, while A2C would require recurrent architectures.

- Replace the aggregate model with individual plant dispatch to test A2C's scalability advantage in high-dimensional settings.

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